1 Planning (15 marks)

Defining the problem (3 marks)

P *m* is the independent variable and *E* is the dependent variable or vary *m* and measure *E*. Do not allow time. [1]

P Keep the temperature change of water constant. Allow two specified temperatures. Do not allow "keep temperature constant". [1]

P Keep the mass or volume of water constant. [1]

Methods of data collection (5 marks)

M Labelled diagram including labelled thermometer with bulb in water and at least one other label. [1]

M Workable circuit diagram to determine *E*: power supply, heater and ammeter and voltmeter, or joulemeter or wattmeter. [1]

M Method to determine change in temperature: measure initial temperature, measure final temperature and subtract, or measure initial temperature and specific temperature change. [1]

M Use balance/scales to measure mass of blocks. [1]

M Stir water (so that metal is in thermal equilibrium). [1]

Method of analysis (2 marks)

A Plot a graph of *E* against *m*. Do not allow log–log graphs. [1]

A *a* = gradient and *b* = *y*-intercept; must be consistent with suggested graph. [1]

Safety considerations (1 mark)

S Precaution linked to hot heater/water, e.g. use gloves **or** use tongs for hot blocks. Do not allow goggles. [1]

Additional detail (4 marks)

D Relevant points might include [4]
1 Method to ensure that e.m.f. of the power supply is constant/current in heater is constant, e.g. adjust variable power supply/variable resistor to ensure p.d./current is constant
2 Keep the starting temperature of water/metal constant
3 Wait for water and metal temperatures to equalise
4 Add insulation to sides of beaker/lid (to prevent energy losses)
5 Use of timer and equation, e.g. $E = Pt = ItV$ for candidate's method
6 Use large temperature change to reduce percentage uncertainty
7 Relationship is valid if the graph is a straight line that does not pass through the origin

Do not allow vague computer methods.

© Cambridge International Examinations 2015

https://xtremepape.rs/